On the bending effects of a printed 4x4 antenna array

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***Abstract*—The bending effects of proximity coupled 4x4 patch arrays with printed patch elements operating in the X-band near 10 GHz is presented. The patch arrays have circular patch elements with inclusive slots residing on the top surface of four layers (10 mil / layer) of DuPont Pyralux AP. The inclusive slots decrease the axial ratio of the antenna and increase the impedance bandwidth. Each array has a microstrip feed network of quarter-wave transformers and T-junction power dividers in the middle of the four layer stackup. The microstrip lines are proximity (or capacitively) coupled to the patch elements. A ground plane on the bottom surface of the four layer stackup provides surface agnostic behavior. The patch elements and microstrip feed network are printed with DuPont CB028 conductive ink using a liquid dispense technique. For the purpose of comparison to PCB processed antennas, 4x4 patch arrays with ½ oz. electroplated copper for the patch elements and microstrip feed network are also reported. The gain, 3 dB beamwidth, and VSWR bandwidth behavior have been studied for uniaxial bending of the two arrays for radii of curvature down to a 1 inch. The nominal (i.e., no bending) measured far-field gain of the arrays is 14.5 dBi (printed) and 14.7 dBi (copper). The printed and copper arrays behave very similarly with changes in gain of 9.3 dB (printed) and 8.3 dB (copper) for bending down to 1 inch in an axial direction and changes in gain of 2.6 dB (printed) and 2.8 dB (copper) for bending down to 1 inch in a circumferential direction.**

Keywords—additive manufacturing, antenna bending, conformal antenna, patch antenna, printed antenna.

I. Introduction

Additive manufacturing of 3D printed parts is becoming increasingly popular in the aerospace industry for replacement of flyaway parts on aircraft. 3D printed parts have low cost, weight, and manufacturing time, while still providing high structural integrity. Similarly, flexible hybrid electronics (FHE) realized by additive manufacturing can be utilized by the aerospace industry for areas such as real time structural health monitoring, diagnostic testing on the factory floor, and improving the in-cabin passenger experience. Printed electronics inks can be dispensed by contact (e.g., liquid dispense, screen print) and non-contact (e.g., aerosol jet dispense, ink-jet dispense) printing techniques.

Conventional PCB processing of antennas uses subtractive methods (e.g., milling, wet etching, and laser ablation) of copper on a circuit board to define feed networks and antenna elements. Subtractive methods, however, have high material waste and are time consuming, resulting in high manufacturing cost [1]. Additive printing methods based on dispensing printing techniques, on the other hand, have negligible material waste and have fast printing speeds, resulting in low manufacturing cost [1].

There have been a number of printed antennas [2]–[10] reported in recent years. Printed conductive inks are advantageous for conformal antennas as they have higher flexibility than bulk copper and can be deposited using a number of available printing tools; on the other hand, they have higher electrical resistivity than bulk copper. For example, the electrical resistivity of DuPont CB028, a silver-based conductive ink, has been reported as 220 nΩ.m [11], which is 14x higher than bulk silver. The surface (or sheet) resistance of a patch antenna is proportional to the electrical resistivity or $R_S \propto \rho$. The loss resistance of a patch antenna is proportional to the surface resistance or $R_L \propto R_S$. The far-field antenna gain of a patch antenna is related to the loss resistance as

$$G \propto \eta_{rad} = \frac{1}{1 + \frac{R_L}{R_{Rad}}}, \tag{1}$$

where R_{Rad} is the radiation resistance of the patch antenna.

The antenna gain of a printed patch antenna is limited by the electrical resistivity of the conductive ink; however, the decrease in gain can be offset by increasing the film thickness.

A microstrip feed network is commonly used for feeding patch antenna arrays. A patch antenna fed by a microstrip line can have one of three feed types: inset feed, proximity coupled feed, or aperture coupled feed [12]. The proximity coupled patch antenna typically has higher impedance bandwidth than the inset feed as the impedance of the antenna can be matched by adjusting both the length of the feeding stub and the patch width-to-line width ratio [12]. Pozar reported the first proximity coupled fed patch antenna [13] and afterwards reported an impedance bandwidth of 13% [14]. A number of methods have been used to improve the performance of coupled patch antennas [15]–[22].

There have been several experimental [23]–[26] and analytical [27]–[30] studies on the bending behavior of planar arrays as well as adaptive approaches for beam compensation [31], [32]; however, to the authors knowledge there has not been an adequate study on the bending behavior of proximity coupled planar arrays. Furthermore, there has not been a study of the performance of additively manufactured printed in comparison to conventional PCB processed copper planar arrays.

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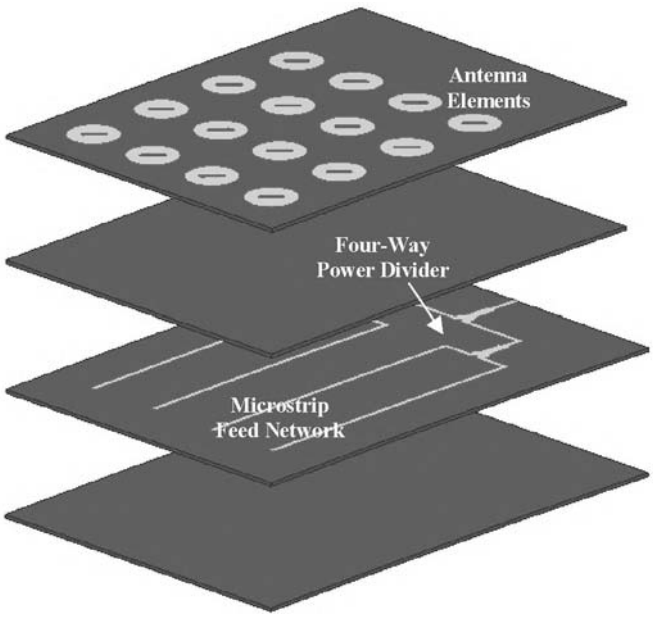

Fig. 1. Perspective view of printed 4x4 patch antenna array with antenna elements, microstrip feed network, and ground plane (not shown).

In this paper, an array consisting of proximity coupled patch antenna elements with inclusive slots above a microstrip feed network and a lower ground plane is described. The inclusive slots decrease the axial ratio for circular polarization and improve the impedance bandwidth [33].

In this paper, an array consisting of proximity coupled patch antenna elements with inclusive slots above a microstrip feed network and a lower ground plane is described. The inclusive slots decrease the axial ratio for circular polarization and improve the impedance bandwidth [33]. The bending effects of printed 4x4 antenna arrays with DuPont CB028 conductive ink antenna elements and copper 4x4 antenna arrays with ½ oz. electroplated copper will be evaluated in the X-band at 10 GHz. The antenna arrays are uniaxially bent in axial (axis along length of microstrip feed line length) and circumferential (axis perpendicular to microstrip feed line length) directions to radii of curvature down to 1 inch to determine the bending effects on the far-field antenna gain, half-power beamwidth, and 2:1 VSWR bandwidth. The following sections describe the design, fabrication, and experimental characterization of printed and copper 4x4 antenna arrays.

II. Antenna Design

In the earlier years of development of patch antennas, transmission line and cavity models were developed for their analysis [12]. The transmission line model is a distributed element model with an input impedance proportional to the patch width or $Z_{in} \propto \tan \beta W$ [12]. The cavity model, on the other hand, is a lumped element model (LEM) that assumes the antenna behaves like a narrow-band resonator with an input impedance equivalent to a parallel RLC circuit [33]. The cavity model places boundary conditions on the patch antenna such that E-fields are only allowed to propagate in the z-direction or $\hat{z} \times [\vec{E}^r + \vec{E}^i] = Z_s[\hat{z} \times \vec{J}_s]$ [34]. The cavity model is a simplistic and accurate model to describe the patch antenna and will therefore be used to analyze the 4x4 antenna arrays.

A. Lumped Element Model

The radius of a circular antenna element is related to the electrical operating wavelength as $b = 1.841\lambda/2\pi$ [35] and the area is related to the radius as $A = \pi b^2$. However, the area does not account for fringing effects along the periphery of the antenna element. The effective antenna radius accounting for the fringe fields is [35]

$$b_e = b\sqrt{1 + \frac{2d}{\pi b \varepsilon_r}\left\{\ln\frac{\pi b}{2d} + 1.7726\right\}}, \tag{2}$$

where d is the distance between the antenna element and the lower ground plane, and ε_r is the relative permittivity of the dielectric medium below the antenna element.

The capacitance of the circular antenna element is

$$C = \frac{\varepsilon_o \varepsilon_r}{d}\int_0^{2\pi}\int_0^{b_e} r dr d\theta, \tag{3}$$

where ε_o is the permittivity of free-space.

The electrical resonant frequency of the circular antenna element is given as

$$f_o = \frac{1}{2\pi\sqrt{LC}} = \frac{c}{\lambda\sqrt{\varepsilon_r}}, \tag{4}$$

where L is the inductance of the circular antenna element derived as $L = \mu_o d/\pi(1.841)^2$ [35] and λ is the electrical operating wavelength in free-space.

The electrical quality factor Q of the antenna element is related to the radiation quality factor Q_r, surface wave quality factor Q_{sw}, dielectric quality factor Q_d, and conductor quality factor Q_c as $Q^{-1} = {Q_r}^{-1} + {Q_{sw}}^{-1} + {Q_d}^{-1} + {Q_c}^{-1}$ [36]. The losses associated with surface waves, dielectric, and conductor are generally small in comparison to the radiation loss such that $Q \approx Q_r$. The electrical quality factor of the circular antenna element is derived as [37]

$$Q = \frac{1.841\pi b R_{rad}\sqrt{\varepsilon_r}}{d Z_o}, \tag{5}$$

where Z_o is the impedance of free-space.

The expression in (5) agrees with Balanis' assessment of patch antennas using a volumetric relationship to determine $Q \propto \sqrt{\varepsilon_r}$ [38]. The fractional bandwidth of the antenna element, which is related to the VSWR bandwidth, is inversely proportional to the quality factor as $\Delta f/f_o = 1/Q$. Planar patch antennas are inherently narrow-band (i.e., high Q factor) due to their small size (i.e., thickness) as predicted by Chu [39] and later validated experimentally by Sievenpiper *et al.* [40].

B. Antenna Elements and Feeding

Ansys HFSS, a finite element method (FEM) numerical solver, is used to develop a model of the 4x4 array. The 4x4 array with its various layers is illustrated in Fig. 1. The array has overall dimensions of 80.6 mm x 55.8 mm. The array is composed of circular antenna elements on the surface of four layers (10 mil / layer) of DuPont Pyralux AP, a flexible polyimide, with ε_r = 3.4 and $\tan\delta$ = 0.002. A 60 GHz patch antenna [41] has previously been reported with DuPont Pyralux

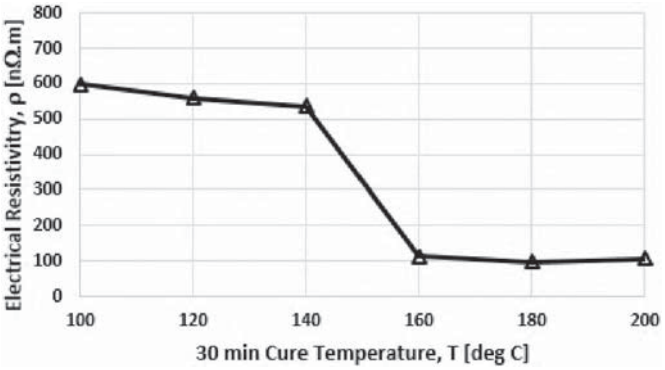

Fig. 2. Measured electrical resistivity of DuPont CB028 conductive ink.

TK, which is similar to DuPont Pyralux AP, but with lower moisture absorption. The array herein is designed to electrically resonate near 10 GHz. The FEM model for a single circular patch antenna element with an antenna radius of 4.15 mm predicts a resonant frequency of 10.5 GHz. The addition of a quarter-wave slot with a length of 4 mm and width of 0.5 mm decreases the predicted resonance of the antenna element to 10.3 GHz. The width of the slot is numerically determined such as to maximize coupling between the antenna element and the microstrip feed network.

The microstrip feed network has a four-way power divider with two stages: stage one is a single quarter-wave transformer [42] fed into a T-junction power divider [43] and stage two is two duplicate quarter-wave transformers fed into two duplicate T-junction power dividers. Each quarter-wave transformer is designed such as to match a 50Ω input into a 25Ω output using a quarter-wave line with an impedance of 35.3Ω. Each T-junction power divider is designed to match a 25Ω input into two 50Ω outputs. The insertion loss at 10 GHz through a four-way power divider used for the 4x4 array is 6.6 dB and the 2:1 VSWR bandwidth is 6.6 GHz.

The center to center spacing of antenna elements along the length of the microstrip feed network is 14.45 mm and the center to center spacing of antenna elements perpendicular to the microstrip feed network is 12.45 mm.

The ideal antenna gain of the 4x4 antenna array is given as [44]

$$G_{ideal} = \eta_{ant} \frac{4\pi A_p}{\lambda^2}, \tag{6}$$

where η_{ant} is the antenna efficiency and A_p is the area of the aperture.

The area of the aperture is related to the length and width of the aperture as $A_p = L_p \times W_p$. Furthermore, the length and width of the aperture are related to the number of elements and element spacing in their directions as $L_p = N_L d_L$ and $W_p = N_W d_W$, respectively. The antenna efficiency is related to the reflection coefficient Γ and the radiation efficiency as $\eta_{ant} = (1 - |\Gamma|^2)\eta_{rad}$. The antenna gain and 3 dB beamwidth are, in part, determined by the VSWR bandwidth, which is related to the reflection coefficient.

The ideal antenna gain for a 4x4 array with an antenna element radius of 4.15 mm, a length of 4 elements spaced 14.45 mm apart, and width of 4 elements spaced 12.45 mm apart is 22.5 dBi. However, the ideal gain does not take into account the losses due to the microstrip lines and power divider (i.e., microstrip feed network) as well as radiation from the slots. The loss through the four-way power divider at 10 GHz is 0.65 dB and the microstrip line loss is 0.12 dB/cm for a total is 6.3 dB for the power divider and 47 cm long microstrip feed line. The realizable gain with the four-way power divider and microstrip line losses as well as assuming 90% antenna efficiency is 15.7 dBi.

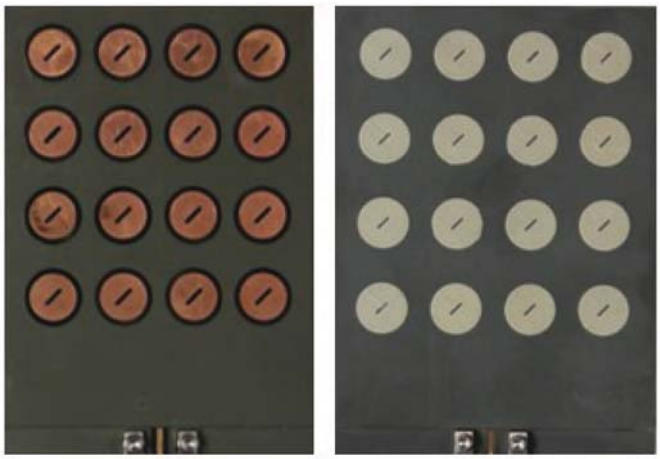
Fig. 3. Fabricated copper and printed 4x4 antenna arrays.

III. Antenna Fabrication

The copper and printed 4x4 arrays are fabricated using both conventional subtractive and additive printing methods. The processes for fabricating the arrays are described in the following sections.

A. Conductive Ink Evaluation

There electrical resistivity of conductive inks are known to be strongly dependent on their curing conditions including: temperature, air flow, humidity, and time. In this study DuPont CB028 conductive ink was dispensed on DuPont Pyralux AP using an nScrypt 3Dn-300 printer. Cloverleaf structures were selected to characterize the electrical resistivity of the conductive ink. The van der Pauw method is used to determine the electrical resistivity of the ink by first applying a voltage across two adjacent electrodes then measuring the current through two opposite electrodes as shown in Fig. 2. The electrical resistivity is determined by the measured resistance and thickness profile of the ink.

The electrical resistivity of DuPont CB028 as a function of curing temperature for 30 min cure times is shown in Fig. 2. The electrical resistivity of DuPont CB028 cured at 160°C for 30 min is 110 nΩ.m, which is 6x higher than bulk silver. The results are 2x better than those reported in literature [11]. The use of liquid dispense printing allows silver flakes to be dispensed on the substrate more readily than screen printing where the mesh tends to block silver flakes. The electrical resistivity of DuPont CB028 is constant above 160°C due to complete removal of the binders and solvents.

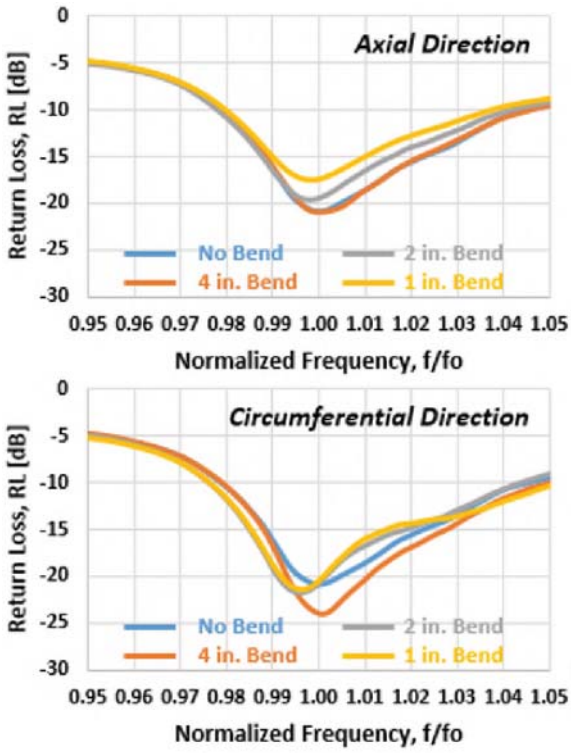

Fig. 4. Measured return loss of printed 4x4 patch antenna array bent down to 1 inch radius in axial and circumferential directions.

B. *Fabrication of Copper and Printed Antenna Arrays*

The copper and printed 4x4 arrays have four layers of 10 mil DuPont Pyralux AP with antenna elements, a microstrip feed network, and a ground plane. The conductive portions of the copper 4x4 array are constructed in ½ oz. electroplated copper. The four layers of DuPont Pyralux AP are processed to laser etch away the unwanted portions of copper using a LPKF ProtoLaser U3 355 nm laser machine. The conductive portions of the printed 4x4 array are constructed by dispensing DuPont CB028 in the desired areas using an nScrypt 3Dn-300 printer. The conductive ink is then baked at 160°C for 30 min in a Thermo Scientific convection oven. The thickness of the antenna elements was measured as 30 μm using a profilometer. The four layers of DuPont Pyralux AP for each array are then aligned for lamination with three layers of 3M 966 adhesive tape placed between the four layers. The adhesive tape has a 2.3 mil thickness with ε_r = 2.9 and tan δ = 0.02 that are accounted for in the FEM model. Finally, the arrays are laminated with a T-Tech multilayer quick press with 48 psi at 70°C for 30 s. The fabricated antenna arrays are shown in Fig. 3.

IV. Experimental Characterization

The following sections will address test setup and measurement results for return loss and far-field gain measurements of the copper and printed 4x4 arrays followed by comparing the measured results to those provided in literature.

A. *Return Loss Measurement*

The copper and printed 4x4 arrays are connected to a high frequency network analyzer to evaluate the return loss characteristics due to uniaxial bending in the axial and circumferential directions. As mentioned earlier, the axial bending direction is the axis along length of microstrip feed line length, whereas the circumferential bending direction is the axis perpendicular to microstrip feed line length.

The nominal (i.e., no bending) 1.8:1 VSWR bandwidths of the copper and printed 4x4 arrays are 2.54 GHz and 2.26 GHz, respectively. The VSWR bandwidths of the two arrays do not show any significant change for uniaxial bending down to a 1 inch radius in either the axial or circumferential direction as shown in Fig. 4.

B. *Far-Field Gain Measurement*

The copper and printed 4x4 arrays are placed in the far-field to determine their radiation patterns under nominal and bending conditions. The Fraunhofer distance, or minimum distance needed to be in the far-field, for a 20 dBi horn antenna with a width of 3.9 inches operating at 10 GHz is 0.65 m. The distance between transmit and receive antennas in the test setup is 4.2 m.

The transmit portion of the test setup for far-field measurement includes a synthesized sweeper and a 20 dBi horn antenna for transmitting a plane wave across free-space at 10 GHz. The receive portion of the test setup includes the antenna under test (i.e., copper or printed 4x4 array), a 30 dB preamplifier, and a power meter with a high frequency power sensor. The setup described can be used to determine a normalized radiation pattern of the antennas. A known reference 20 dBi horn antenna, however, is used in place of the antenna under test to determine the actual gain of the arrays.

The nominal (i.e., no bending) antenna gains for the copper and printed 4x4 arrays are measured as 14.7 dBi and 14.5 dBi, respectively. The FEM model predicted nominal antenna gains of 15.5 dBi (copper) and 15.2 dBi (printed). The measured antenna gains of the copper and printed 4x4 arrays are 0.7 dB lower than the FEM model. The difference between the measured results and FEM model is likely due to factors including: 0.2 dB of connector loss, measurement uncertainty in the test setup, and possible misalignment between layers. Despite the slight differences between the measured results and the FEM model, there appears to be good agreement the measured results and the FEM model, which uses a measured ink thickness of 30 μm and resistivity of 220 nΩ.m for the printed antenna elements and printed microstrip feed network.

The copper and printed 4x4 arrays are scanned to determine radiation patterns for nominal and bending in the axial and circumferential directions. It was determined in the FEM model that the main beam of the antenna array is predicted to be 20 deg off boresight (as shown in Fig. 5) due to an intrinsic phase difference between antenna elements along the length of the microstrip line as a result of serial feeding.

The copper and printed 4x4 arrays had their antenna gains decrease by 8.3 dB and 9.3 dB, respectively, for a 1 inch bend in the axial direction. The 1 dB difference in gain between the two arrays is likely due to misalignment error in the measurement setup. The copper and printed 4x4 arrays had their antenna gains decrease by 2.8 dB and 2.6 dB, respectively, for a 1 inch bend in the circumferential direction. The large change in antenna gain in the axial direction is predicted by the FEM model. The cause of this change in antenna gain for bending in the axial direction is an element to element spacing change of

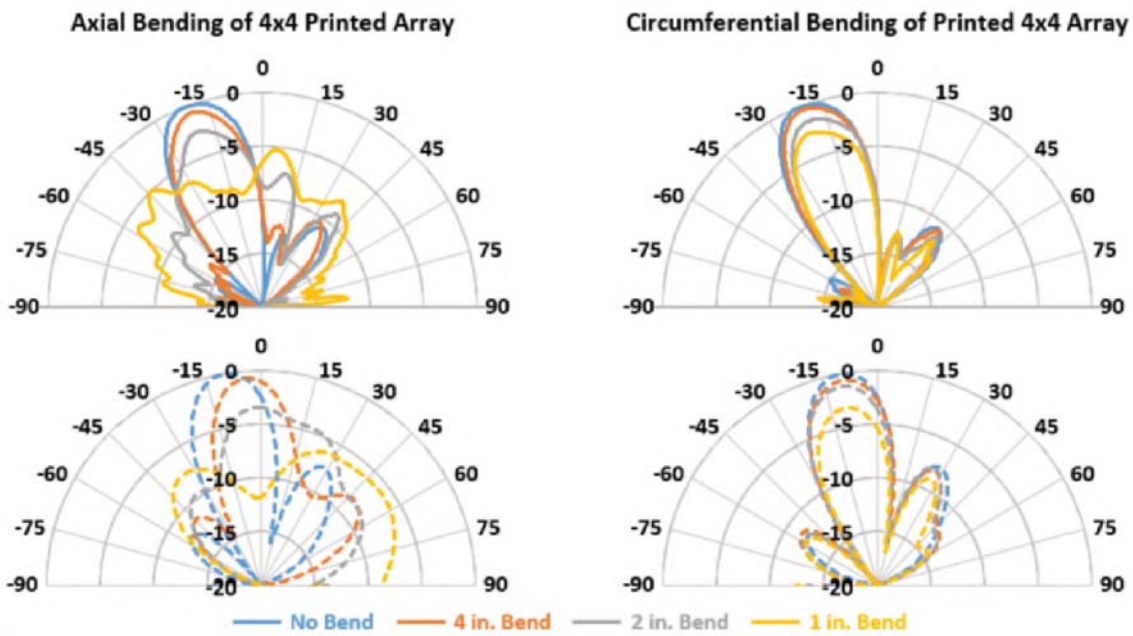

Fig. 5. Normalized measured (top) and FEM (bottom) radiation patterns of printed 4x4 patch antenna array bent down to 1 inch radius in axial and circumferential directions.

4.6 deg in phase (for a 1 inch bend) for elements along the length of the microstrip line, whereas the element to element spacing for elements along the length of the microstrip line did not change for bending in the circumferential direction.

The radiation efficiency of an antenna is a measure of how much power that goes into an antenna is radiated out of the antenna. The nominal radiation efficiencies of the copper and printed arrays were predicted as 91% and 83%, respectively. It was determined through modeling that the radiation efficiencies for the arrays did not change as a function of bending down to a 1 inch bend. Instead, the measured and predicted results show that the radiated power is distributed by the 3 dB (half-power) beamwidth, which is related to the gain as $G \propto 1/\theta_{3dB}$ [45].

The 3 dB beamwidth is experimentally determined from the data acquired from the radiation patterns. The measured 3 dB beamwidth for the copper and printed 4x4 arrays for bending down to 1 inch in the circumferential direction is 23 deg, which is a result of the small change in antenna gain. The 3 dB beamwidth of the arrays is 23 deg for antenna bending down to

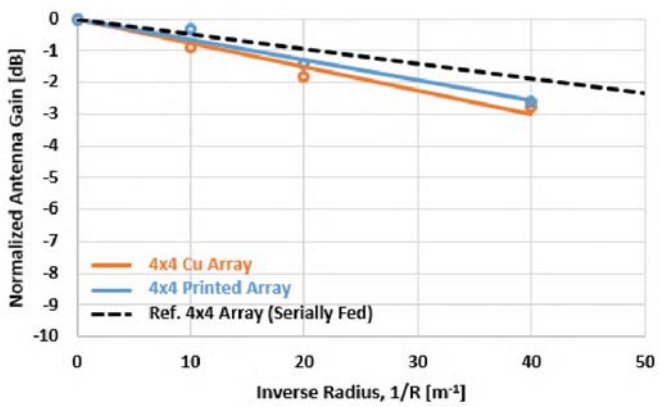

Fig. 6. Normalized antenna gain for copper and printed 4x4 patch antenna arrays in comparison to 4x4 serially fed patch antenna [26].

2 inches in the axial direction; however, the 3 dB beamwidth significantly increases to 104 deg (copper) and 136 deg (printed) for a 1 inch bend, which is a result of a large change in antenna gain. It can be determined that for a serially fed array of proximity coupled patch antennas the antenna gain and 3 dB beamwidth are largely a function of the bending radius for bending in the axial direction. The return loss appears to be fairly constant for bending in either axial or circumferential directions.

C. Comparison to Literature

As mentioned earlier, there have been have been several experimental [23]–[26] studies on the bending behavior of planar arrays. Semkin *et al*. [26] studied the bending behavior of a 4x4 serially fed patch array operating near 59 GHz, which is comparable to the 4x4 aperture coupled patch arrays discussed in this paper. Semkin *et al*. measured the antenna gain of the 4x4 serially fed patch antenna as it was bent in a circumferential direction down to a 6 mm radius. For comparison, the copper and printed 4x4 arrays described in this paper are also bent in a circumferential direction. The normalized gain of the 4x4 serially fed patch antenna and the copper and printed 4x4 arrays as a function of being radius is shown in Fig. 6. The measured gain of the arrays described in this paper match well with the measured gain of the array from Semkin *et al*. The measured gain of the arrays are within 1 dB down to a 1 inch bend.

CONCLUSIONS

The bending of aperture coupled 4x4 antenna arrays has been performed in both axial and circumferential directions to better understand gain, 3 dB beamwidth, and VSWR bandwidth behavior. The nominal antenna gains in the X-band at 10 GHz for copper and printed 4x4 arrays are 14.7 dBi and 14.5 dBi, respectively, which match well. The antenna gain for the copper and printed 4x4 arrays decreased 8.3 dB and 9.3 dB, respectively, for a 1 inch bend in the axial direction. The antenna gain for the arrays decreased 2.8 dB (copper) and 2.6 dB (printed) for a 1 inch bend in the circumferential direction. The

antenna gain decreased significantly more in the axial direction due to a 4.6 deg chance in phase between elements along the length of the microstrip line. The 3 dB beamwidth was fairly constant for bending in circumferential direction and increased significantly for a 1 inch bend in the axial direction due to the phase change described above.

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